

Experiment 24

Title

Write a Prolog Program to suggest a Dieting System based on Disease.

Aim

To implement a Prolog program that suggests a suitable diet based on a given disease.

Objective

To understand how Prolog uses facts and rules to recommend an appropriate diet for different diseases.

Algorithm

Step 1: Define facts for different diseases and their recommended diets.

Step 2: Create a rule to retrieve the diet for a given disease.

Step 3: Enter the disease name through a query.

Step 4: Prolog searches the database for the matching disease.

Step 5: Display the recommended diet.

Program

% Diet Database

diet(heart_disease, [apple, carrot, chicken, fish, almonds]).

diet(diabetes, [apple, carrot, fish, spinach, almonds]).

diet(anemia, [spinach, chicken, fish, almonds]).

diet(obesity, [apple, carrot, fish, spinach]).

% Rule to Suggest Diet

suggest_diet(Disease, Diet) :-

 diet(Disease, Diet).

Sample Input

?- suggest_diet(diabetes, Diet).

?- suggest_diet(obesity, Diet).

Sample Output

```
consult('C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp24.pl').  
compiling C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp24.pl for byte code...  
C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp24.pl compiled, 10 lines read - 1632 bytes written, 5 ms
```

```
(16 ms) yes  
| ?- suggest_diet(diabetes, Diet).
```

```
Diet = [apple,carrot,fish,spinach,almonds]
```

```
(16 ms) yes  
| ?- suggest_diet(obesity, Diet).
```

```
Diet = [apple,carrot,fish,spinach]
```

```
yes
```

Result

Hence, the Prolog program to suggest a **Dieting System based on Disease** was executed successfully and the correct diet recommendations were obtained.